VNet-to-VNet VPN Implementation Using Azure VPN Gateway

Name: Kushagar

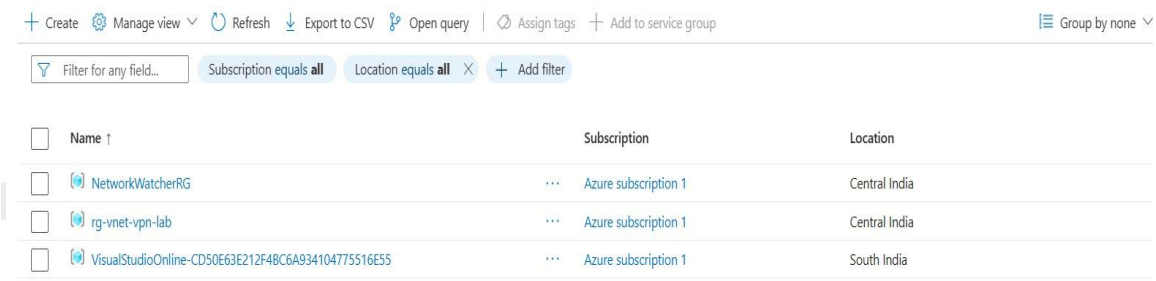
Objective

To create a secure connection between two Azure Virtual Networks using VPN Gateway and verify connectivity using virtual machines.

Detailed Implementation Steps

Step 1: Create Resource Group

Name: rg-vnet-vpn-lab
Region: Central India

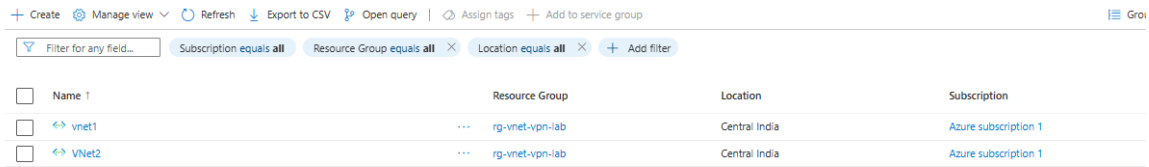


Step 2: Create VNet1

Name: vnet1
Address space: 10.0.0.0/16
Subnet1: 10.0.1.0/24
GatewaySubnet: 10.0.255.0/27

Step 3: Create VNet2

Name: vnet2
Address space: 10.1.0.0/16
Subnet1: 10.1.1.0/24
GatewaySubnet: 10.1.255.0/27



Step 4: Create VPN Gateway for VNet1

Name: vnet1-gateway
Type: VPN
VPN Type: Route-based
SKU: VpnGw1
Public IP: pip-vnet1-gw
Active-active: Disabled
BGP: Disabled

Step 5: Create VPN Gateway for VNet2

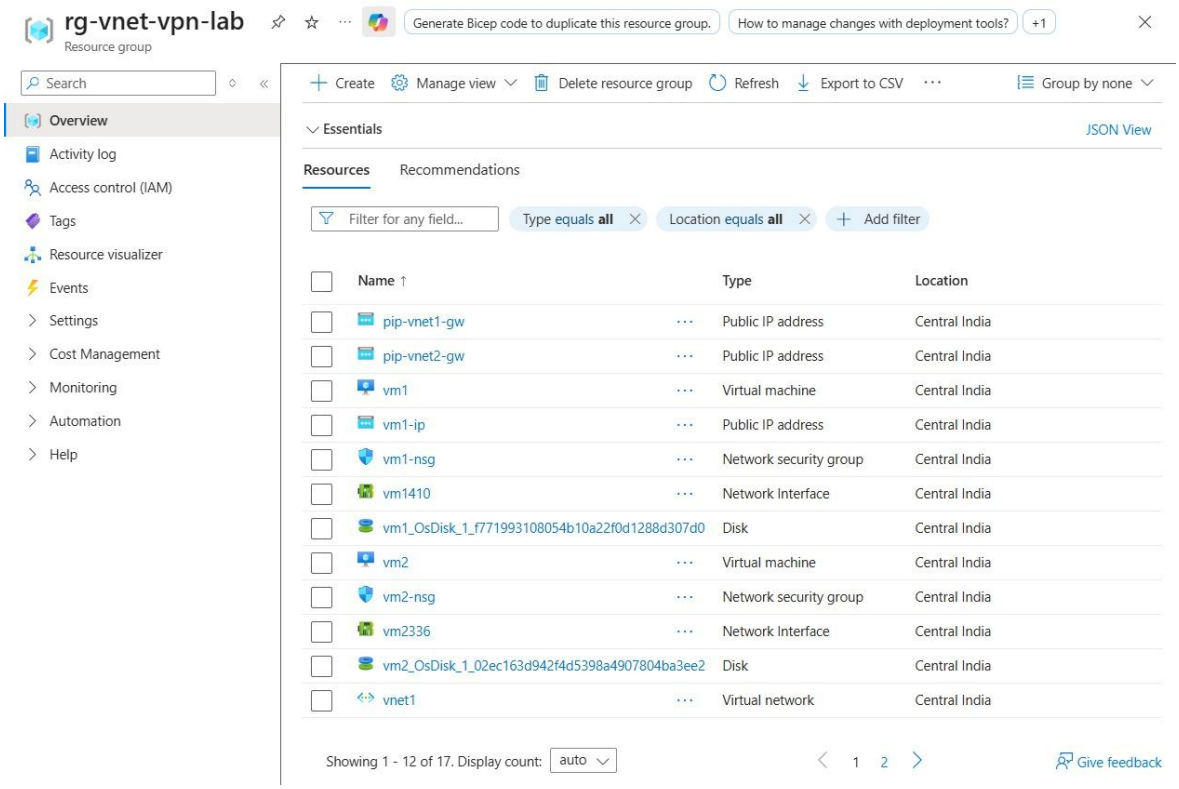
Name: vnet2-gateway
Public IP: pip-vnet2-gw
Same configuration as above

Step 6: Create Connection

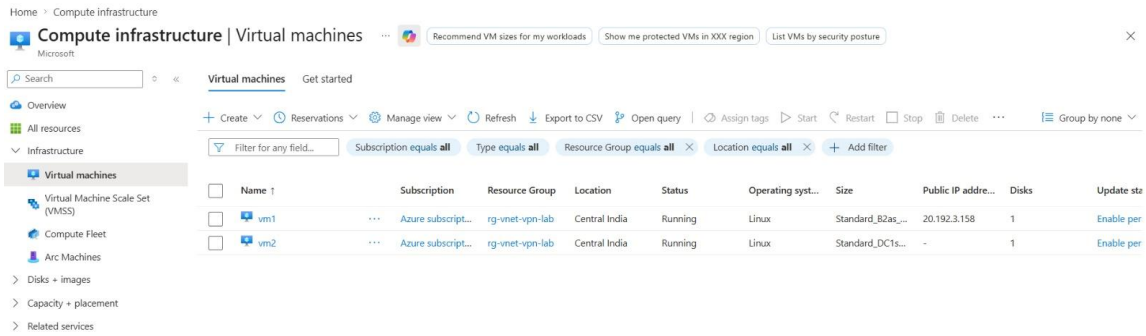
Connection Name: vnet1-to-vnet2
Type: VNet-to-VNet
Shared Key: Azure123

Step 7: Reverse Connection

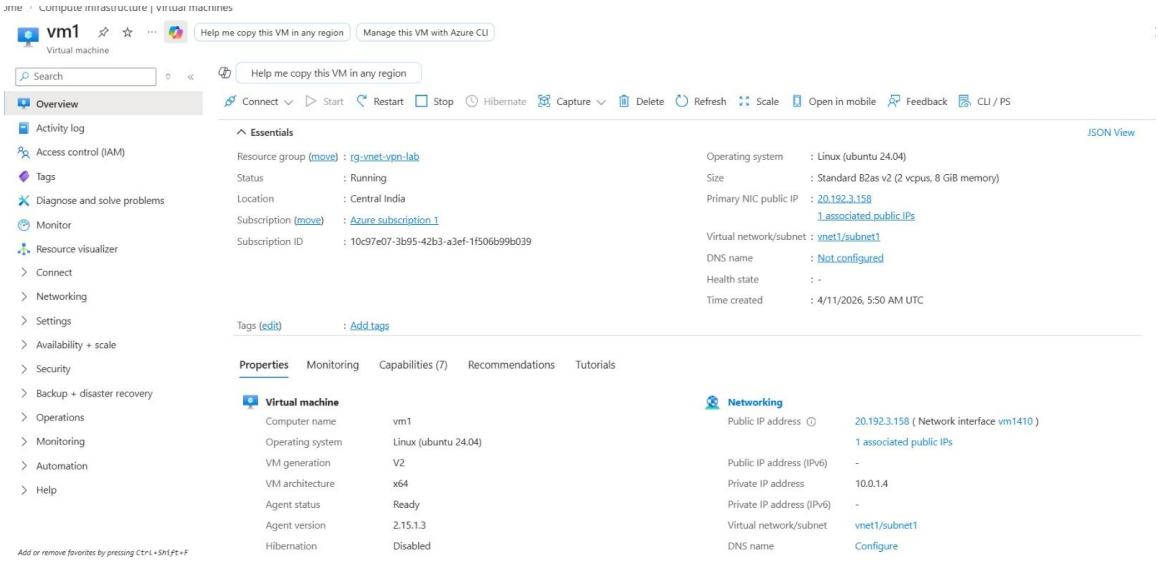
Connection Name: vnet2-to-vnet1
Shared Key: Azure123



Step 8: Create Virtual Machines

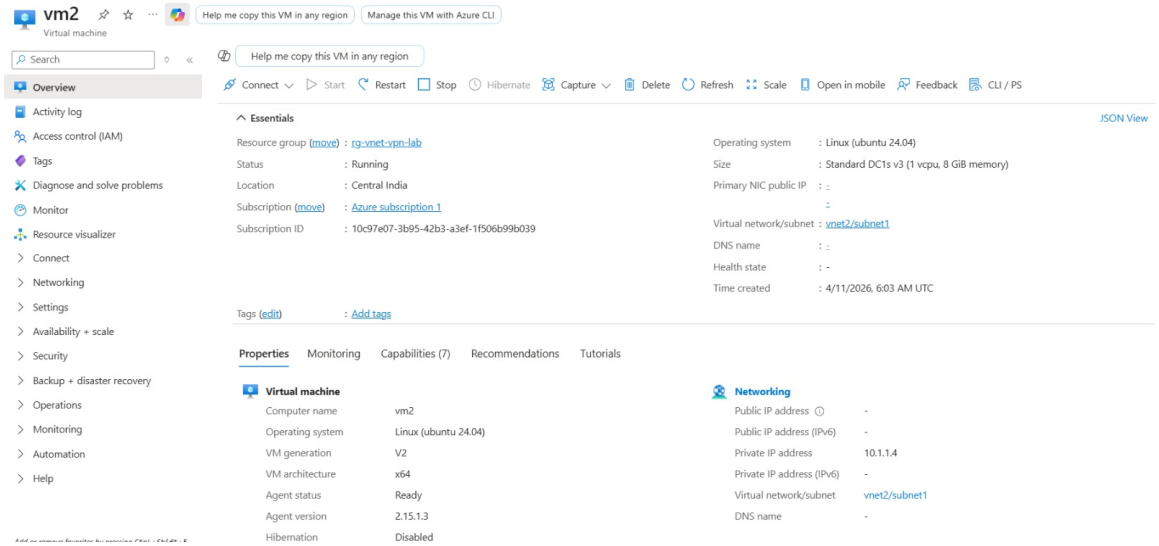


VM1:
Name: vm1
VNet: vnet1
Subnet: subnet1
Public IP: Enabled
Username: azureuser
Password: (user-defined)



VM2:
Name: vm2
VNet: vnet2
Subnet: subnet1
Public IP: None (due to free subscription limit)
Username: azureuser1
Password: (user-defined)

Note: VM2 was created without public IP because Azure free subscription allows limited public IPs.



Step 9: Connectivity Test

From VM1:
ping 10.1.1.4
From VM2:
ping 10.0.1.4

```
Switch to PowerShell  Restart  Manage files  New session  Editor  Web preview  Settings  Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
kush [ ~ ]$ ssh azureuser@20.244.44.135
The authenticity of host '20.244.44.135 (20.244.44.135)' can't be established.
ED25519 key fingerprint is SHA256:+NAFO5lfxXddo8ulOt7btQ2cL93ikIZahSNrygWWpJs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.244.44.135' (ED25519) to the list of known hosts.
azureuser@20.244.44.135's password:
Permission denied, please try again.
azureuser@20.244.44.135's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Apr 11 06:08:38 UTC 2026

  System load:  0.07              Processes:             134
  Usage of /:   5.8% of 28.02GB   Users logged in:       0
  Memory usage: 4%                IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
```

```
azureuser@vm1:~$ ssh azureuser1@10.1.1.4
azureuser1@10.1.1.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Apr 11 06:13:29 UTC 2026

  System load:  0.0               Processes:             125
  Usage of /:   5.7% of 28.02GB   Users logged in:       0
  Memory usage: 4%                IPv4 address for eth0: 10.1.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm1:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=3.02 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=2.20 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=2.51 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=7.63 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=2.09 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=3.91 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=2.15 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=3.13 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=2.23 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=2.05 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=2.18 ms
64 bytes from 10.1.1.4: icmp_seq=12 ttl=64 time=2.05 ms
64 bytes from 10.1.1.4: icmp_seq=13 ttl=64 time=2.11 ms
64 bytes from 10.1.1.4: icmp_seq=14 ttl=64 time=2.14 ms
64 bytes from 10.1.1.4: icmp_seq=15 ttl=64 time=2.32 ms
64 bytes from 10.1.1.4: icmp_seq=16 ttl=64 time=2.36 ms
64 bytes from 10.1.1.4: icmp_seq=17 ttl=64 time=2.20 ms
64 bytes from 10.1.1.4: icmp_seq=18 ttl=64 time=2.13 ms
64 bytes from 10.1.1.4: icmp_seq=19 ttl=64 time=1.95 ms
64 bytes from 10.1.1.4: icmp_seq=20 ttl=64 time=2.07 ms
64 bytes from 10.1.1.4: icmp_seq=21 ttl=64 time=4.46 ms
64 bytes from 10.1.1.4: icmp_seq=22 ttl=64 time=2.73 ms
64 bytes from 10.1.1.4: icmp_seq=23 ttl=64 time=2.07 ms
64 bytes from 10.1.1.4: icmp_seq=24 ttl=64 time=2.28 ms
64 bytes from 10.1.1.4: icmp_seq=25 ttl=64 time=2.56 ms
64 bytes from 10.1.1.4: icmp_seq=26 ttl=64 time=2.01 ms
64 bytes from 10.1.1.4: icmp_seq=27 ttl=64 time=2.07 ms
64 bytes from 10.1.1.4: icmp_seq=28 ttl=64 time=2.35 ms
64 bytes from 10.1.1.4: icmp_seq=29 ttl=64 time=2.53 ms
64 bytes from 10.1.1.4: icmp_seq=30 ttl=64 time=12.4 ms
64 bytes from 10.1.1.4: icmp_seq=31 ttl=64 time=2.21 ms
64 bytes from 10.1.1.4: icmp_seq=32 ttl=64 time=1.98 ms
64 bytes from 10.1.1.4: icmp_seq=33 ttl=64 time=8.91 ms
64 bytes from 10.1.1.4: icmp_seq=34 ttl=64 time=1.88 ms
^C
--- 10.1.1.4 ping statistics ---
34 packets transmitted, 34 received, 0% packet loss, time 33047ms
```

Conclusion

The VNet-to-VNet VPN was successfully created and verified using ping between virtual machines.